Ayoub ABBANI

Software Engineer ~ 4 years of experience

+32472701239 | abbaniayoub@gmail.com | Wavre , Belgium | Portfolio | MyGithub | MyLinkedin

Skills

Fluency in English & French

Core Areas of Expertise:

Software Development.
Embedded Software.
Software Testing.
Deep Learning/Computer Vision.
API Integration Automation & CI/CD Pipelines.
SDLC & Agile/Scrum/Waterfall Methodologies.

Technical Skills:

Languages & Frameworks: C++11/14/17, Qt5/QML, Python & Flask HTML/CSS/JavaScript, Electron, NodeJs, RESTful APIs, JSON, XML.
Scripting & Tools: CLI, Shell Scripting, Docker, AWS, SSH, VNC, Mobaxterm vSphere.
Build & Testing: CMake, GMock/Gtest, Pytest.
Databases: SQL, MySQL, MariaDB, MongoDB.
Version Control & CI/CD: Git, Bitbucket, GitHub, Jenkins.
IDEs: Qt Creator, VS Code.
Environments: Windows, Linux.

Experience

EUROCONTROL

Brussels, Belgium
Jun. 2023 - Present

R&D Software Engineer | Surveillance & ATM (Contractor via Akkodis)
1 year, 5 months

Key Responsibilities and Achievements : Contributed to the full lifecycle development and maintenance of the SASS-C software within Radar and Air Traffic Management Surveillance, Delivering robust algorithms for interoperable, High-level functionality to civil and military Air Navigation Service Providers (ANSPs), R&D and the ATM industry.

- **Re-engineering of Screening file Viewer (VERIF 24.04**): Implemented new features and refactored legacy code, including the upgrade of Qt 5.15.3 with QML and migration from C++03 to C++11/C++14 standards.
- **Radar Data Processing (VERIF 24.04.1):** Developed a grid screening feature in VERIF 24.04.1 to support radar GEOTIFF file processing and enhancing trajectory reconstruction in OTR.
- **Cloud-Based Data Management EDIS (VERIF 24.10):** Delivered EDIS, a cloud-based web/desktop module within VERIF 24.10 with **Electron and Flask**, to manage and integrate Radar data in VERIF-DB efficiently.
- **Infrastructure & DevOps:** Configured Oracle Linux (7, 8, 9) environments for VERIF installation, execution, and validation testing, Configured virtual GPU drivers to ensure compatibility and contributed to CI/CD Jenkins pipeline.
- **Quality Assurance & Technical support**: drafted technical specifications, automated tests using python, and documentation, Provided Tier 3 support by investigating incidents via **Mantis** and delivering prompt feedback to the users.

Tech & Tools: C++11/14, Qt5/ QML, Python3, Flask, Electron, HTML CSS JS, Shell Scripting, CMake, mariaDB5, VNC, Git & Bitbucket, Docker, Oracle Linux, CI/CD Jenkins, Mantis & Jira.

STELLANTIS

Tangier, Morocco | Île-de-France, France
Jan. 2022 - Jun. 2023

Software Engineer | XIL ePowertrain (Contractor via Expleo)
1 year, 6 months

Contributed to the development and optimization of a software tool for Automated validation HIL and SIL on SCALEXIO benches. Enabled full Integration with dSPACE, diagnostic and calibration tools, enhancing ePowertrain/ECU validation process.

Key Responsibilities and Achievements:

- **XiL Framework Development:** Built +50 custom library within AutomationDesk using **ASAM XiL API and Python**, enabling seamless API-based communication with tools such as ControlDesk and VEOS to support automated testing processes.
- **Diagnostics Library Integration:** Developed libraries for integration with Diagalyser, automating OBD testing (DTCs, routines, DIDs, services) in compliance with ISO 14229, improving fault diagnosis efficiency.
- **ECU Calibration :** Developed API calls & requests to manage AUTOSAR measurement data and ECU calibration with COM-API and ETAS INCA, enhancing ECU Automated calibration.
- **Automated Test Deployment:** Deployed automated testing across multiple ECUs and ePowertrain projects on SCALEXIO benches using a software tool developed in C++11 and Python, enhancing efficiency and consistency in validation testing.
- **Global Collaboration:** Collaborated across teams (Romania, France, Morocco) to enhance functional testing standards and code reviews in an agile setting.

Tech & Tools: Python, C++11, ASAM XiL API, Qt5, Git/GitHub, ControlDesk, VEOS, AutomationDesk dSPACE, INCA & MDA, DiagAlyser,CAN & UDS, COM-API.

STELLANTIS

Casablanca, Morocco
Feb. 2021 - Jan .2022

Software Integration & Validation Engineer
1 year

as a SW Integrator & In Vehicle SW Tester in ICE Projects serial life

Key Responsibilities and Achievements:

- Perform SW Integration/upgrade in vehicles ECUs starting from a given reference electronic configuration
- Executed +30 test plans in HIL SCALEXIO Bench's in CAN/OBD/EE Faults/RCW specifications
- Develop CAPL script to perform CAN/LIN Network/Sensors & Actuators/ADAS Test using diagnostic tools and methods
- Identify E/E issue and get logs following a defined procedure.
- Provide Root Cause analysis and document issue with description readout, measurements, and log files

Tech & Tools: VECTOR CAN/LIN Interface, CANalyzer, CAPL, DiagAlyser, ETAS ECU & BusInterface , INCA/MDA dSPACE: ControlDesk, HIL Bench (SCALEXIO), Python Scripting, UDS/ CAN Network.

Education

Sultan Moulay Salimane University - FST
Engineer's degree in Mechatronics Engineering
Beni Mellal, Morocco
Sep. 2018 - Jun. 2021

ALTEN National Software Hackathon (AMSA): **Won third prize** for developing an AD/ADAS feature with Deep Learning C++ on Raspberry Pi4. *Dec. 2020*
Showcase: Prototype / Final Result.

Mohammed V University in Rabat - FS
Bachelor of Science - BS in Physics
Rabat, Morocco
Sep. 2015 - Jun. 2018